

Auto Ticket Classification using Flow Designer

Problem Statement

The school IT helpdesk receives multiple incident requests daily from students and teachers, such as Wi-Fi issues, projector failures, password problems, and slow computers.

Currently, IT staff manually reviews each request and assigns a category, which is time-consuming and inefficient.

This project automates ticket classification using ServiceNow Flow Designer by analysing keywords in the incident's short description and description fields.

Objective

- Automatically classify incidents at the time of creation.
- Reduce manual effort for IT agents.
- Improve ticket routing efficiency.
- Implement a no-code, easily maintainable solution.

PHASE 1: Requirement Analysis & Planning

The school IT helpdesk receives multiple support requests daily from students and teachers related to common issues such as network connectivity, hardware failures, account access problems, and system performance issues.

Currently, IT staff manually review each ticket to:

- Identify the issue type.
- Select the correct category and subcategory.
- Inform the caller about ticket creation.

This manual process is time-consuming, error-prone, and not scalable as the number of requests increases.

Activity : 1 Business Use Case.

Business Requirements

The system must:

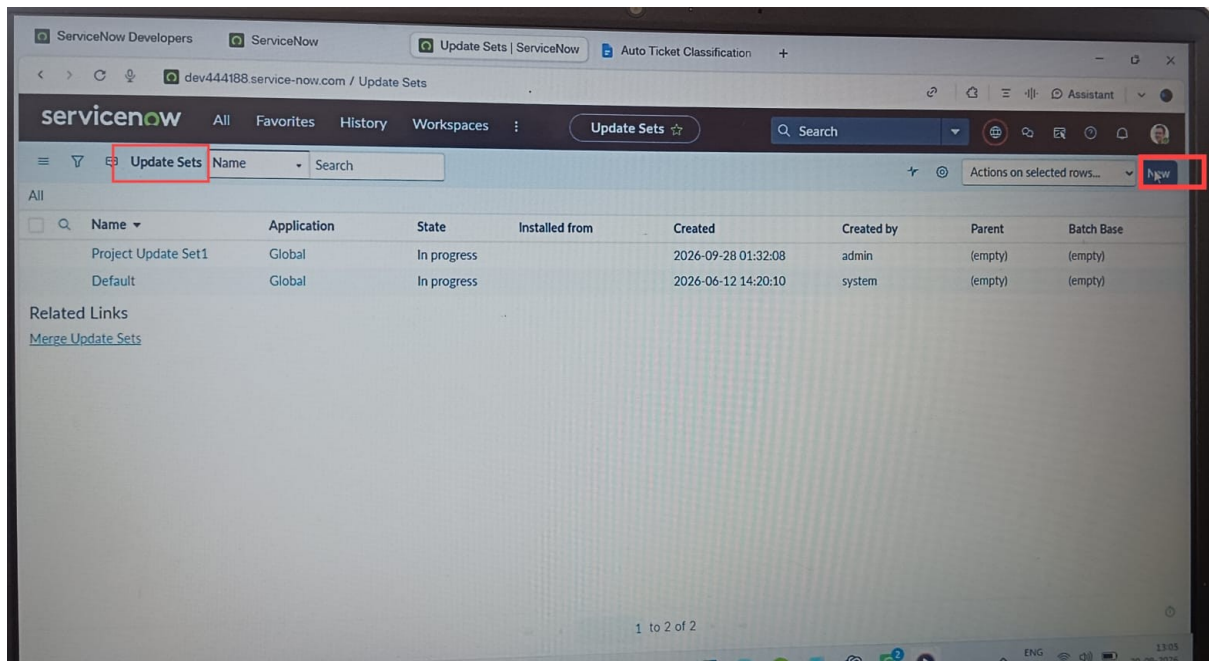
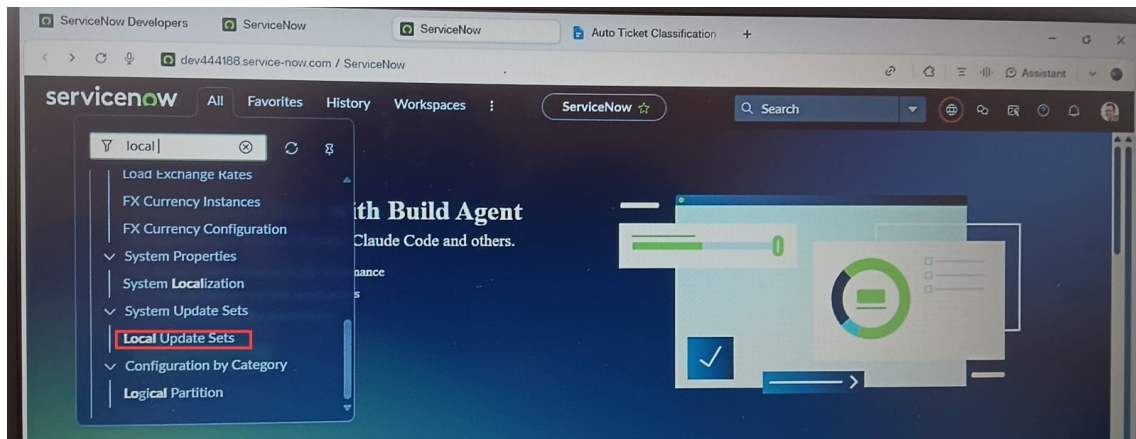
1. Automatically classify IT tickets based on the issue description.
2. Assign both Category and Subcategory without manual intervention.
3. Support dependent choice logic between Category and Subcategory.
4. Send an automated email notification to the caller upon ticket creation.
5. Store ticket information in a structured and standardized format
6. Ensure easy maintenance and future scalability.

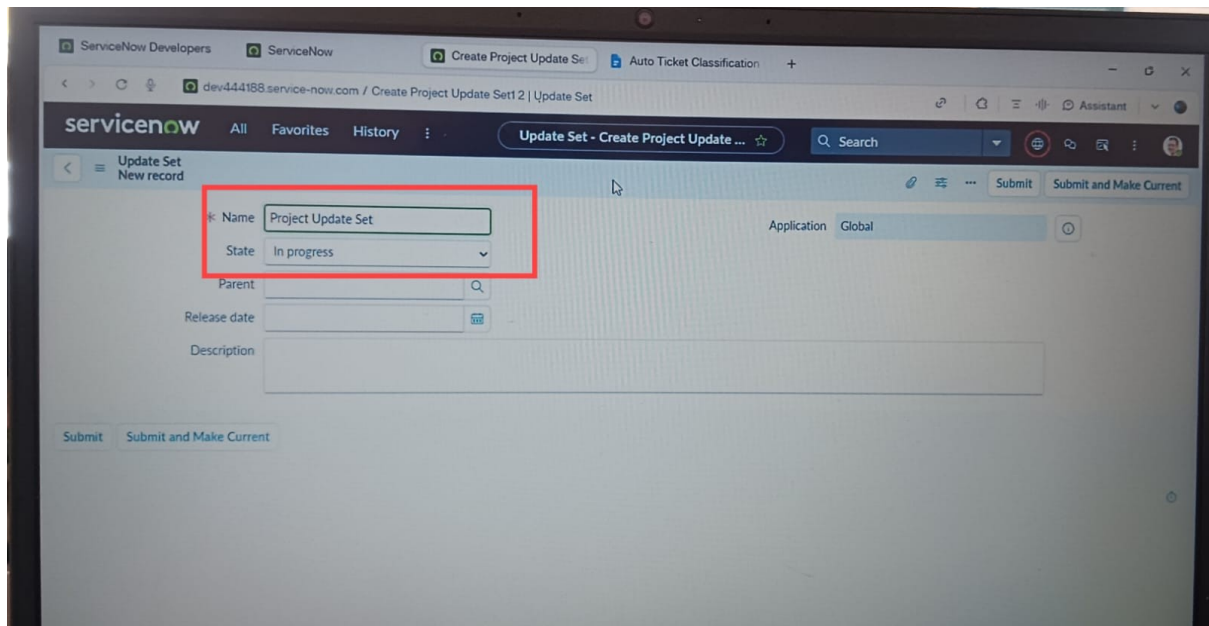
Activity : 2 Creation of New Update Set.

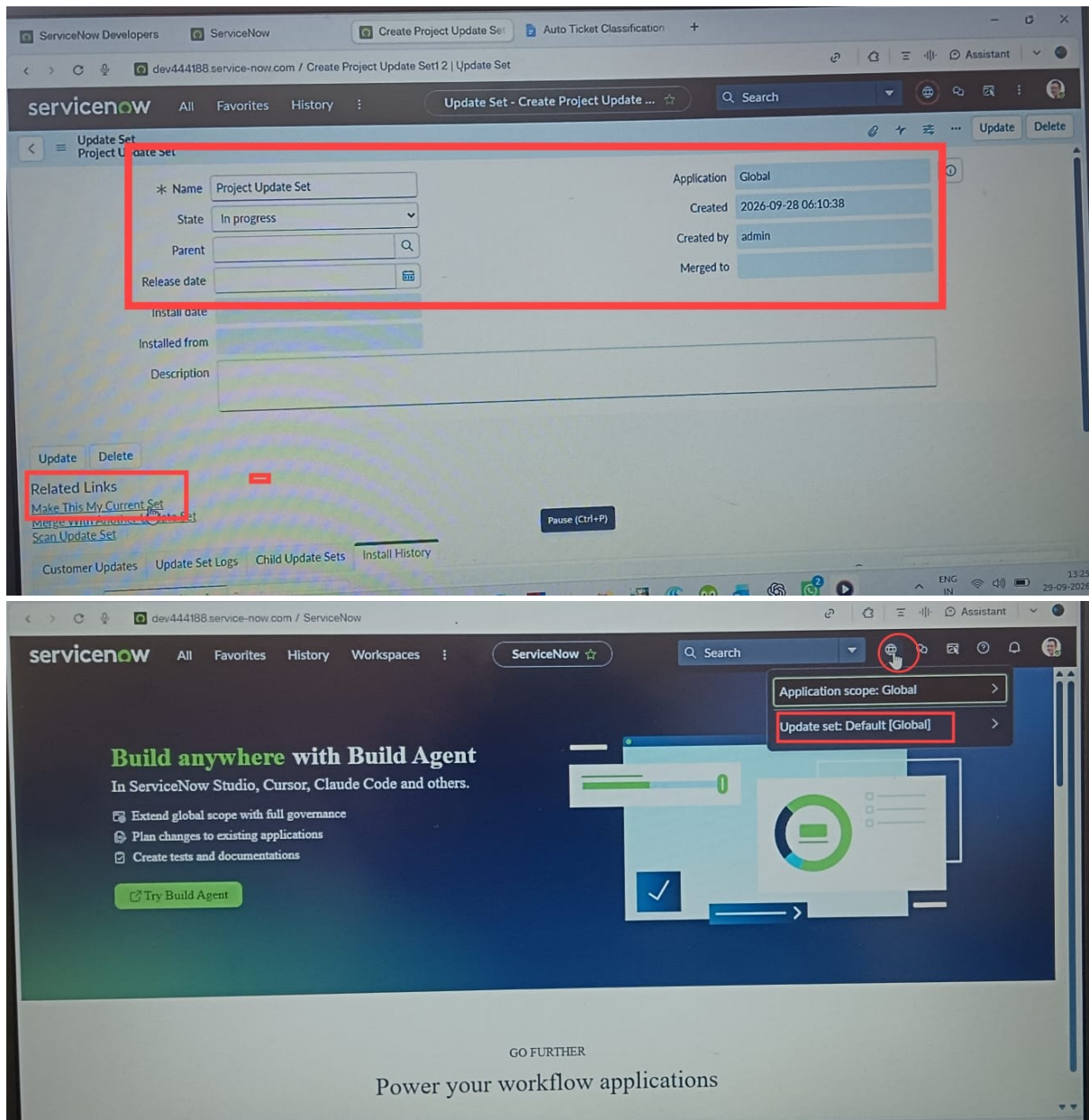
Navigation Steps :

- Go to All menus → Search for Update set.
- Under System Update Sets → Select Local Update Sets.
- Click the New button on the top right corner of the page.
- Name : **Project Update Set.**
- State : **In progress.**
- Application : **Global.**
- Then **Save the Form.**
- Under Related links Click on the **Make This My Current Set.**
- Then the Created Update Set will appear as Current.

Activity : 3 Navigation steps in Picture View.







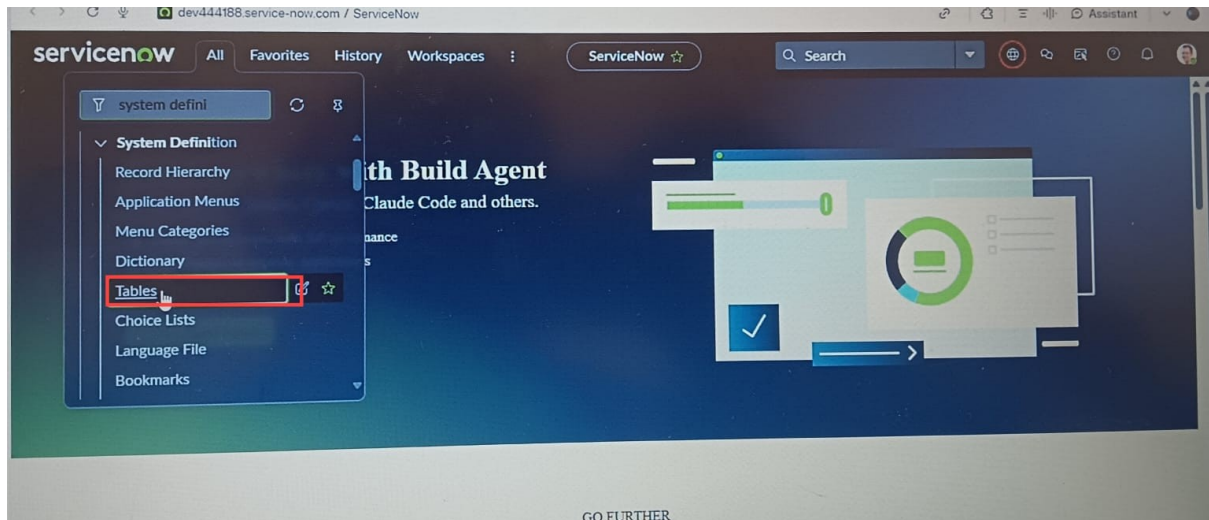
Phase :- 2 Backend Development & Configuration

Activity 1 : Creation of Custom Table to Store Ticket Records.

Steps to Create Custom Table

- Open All menus → Search for Tables.
- Under **System Definition** → Click **Tables**.
- On the top right corner Click **New** button.
- Label : **Incident WorkFlow**.
- Uncheck the **Create module**.

- Now **Save** the Form.
- Then Click the **Controls** Suction and Enable the Check box **true Auto Number**.
- Now again **save** the Form.



The screenshot shows the ServiceNow 'Tables' list view. The 'Tables' tab is selected, and the 'Name' column is highlighted. A 'New' button is visible in the top right corner, also highlighted with a red box. The table below lists various system tables with their labels, names, extension details, and update timestamps.

Label	Name	Extends table	Extensible	Updated
Account Subscription Entitlement	account_subscription_entitlement	(empty)	false	2026-06-12 14:52:23
Adaptive Authentication Event	adaptive_auth_event	(empty)	false	2026-06-12 14:41:06
Agent Assist Recommendation	agent_assist_recommendation	Application File	false	2026-06-12 14:35:51
MID Server File	agent_file	(empty)	false	2026-06-12 14:22:34
AIG Access Policy	aig_access_policy	Application File	false	2026-06-12 14:01:27
AIG Access Policy Scope Mapping	aig_access_policy_scope_mapping	Application File	false	2026-06-12 14:01:28
AIG Scope Tool Mapping	aig_scope_tool_mapping	Application File	false	2026-06-12 14:01:26
Record Producer Configuration	aisa_rp_config	Application File	false	2026-06-12 14:39:11
Search Actions	aisa_ui_action	Application File	false	2026-06-12 14:39:11
AI Search ACL Overrides	ais_acl_overrides	Application File	false	2026-06-12 14:03:37
AI Search Active Table Ingestion Tracker	ais_active_table_ingestion_tracker	(empty)	false	2026-06-12 14:03:35
Ais Api Transaction Log	ais_api_transaction_log	(empty)	false	2026-06-12 14:03:39
AI Search Async Genius Result	ais_async_genius_result	(empty)	false	2026-06-12 14:03:39
AI Search Async Request	ais_async_request	(empty)	false	2026-06-12 14:03:40

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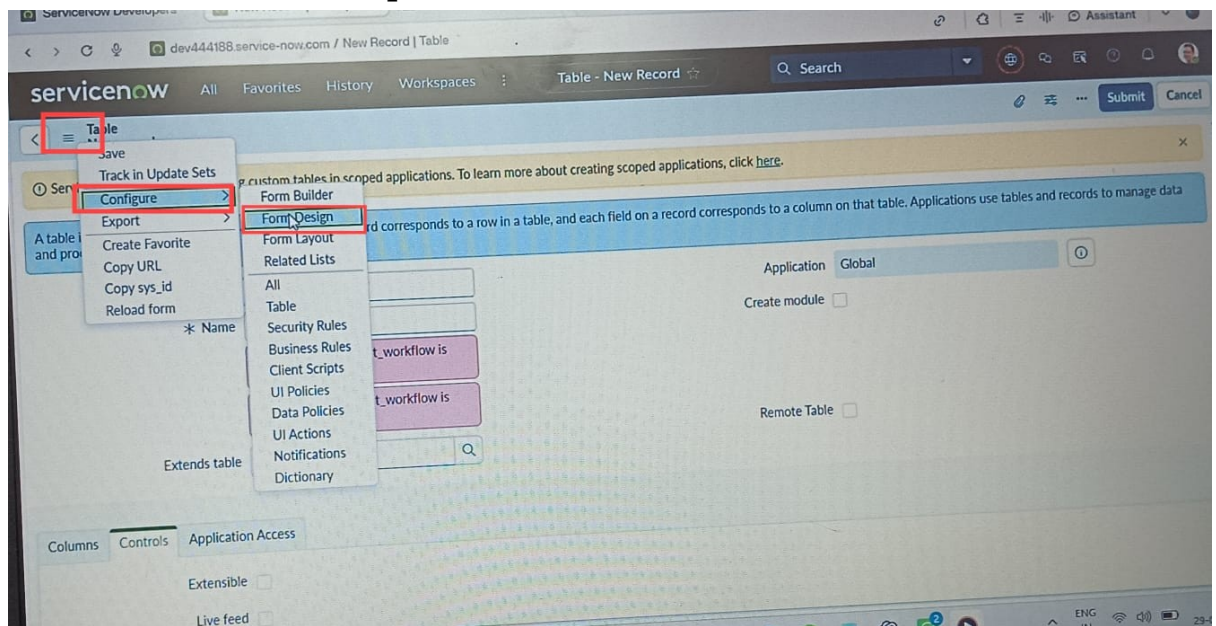
The image consists of two screenshots of the ServiceNow 'Table - New Record' form.

Top Screenshot: The 'Auto-number' section is highlighted with a red box. It includes a checkbox for 'Auto-number' (checked), a 'Prefix' field with the value 'INC', a 'Number' field with the value '1000', and a 'Number of digits' field with the value '7'. Below this, there is a section for 'Security Rules (ACLs)' and a 'Create access controls' checkbox (checked) with a 'User role' field set to 'u_incident_workflow_user'.

Bottom Screenshot: The 'Label' and 'Name' fields are highlighted with a red box. The 'Label' field contains 'Incident WorkFlow' and the 'Name' field contains 'u_incident_workflow'. To the right, there are several configuration options: 'Application' (Global), 'Create module' (checked), 'Create mobile module' (checked), 'Add module to menu' (set to '-- Create new --'), 'New menu name' (Incident WorkFlow), and 'Remote Table' (unchecked).

- Then Save the Form and Under Related Links Click **Show Form**.
- Now Open the Configure On clicking the Context Menu and then Open the **Form Designer**.

- Then Create Fields as Required.

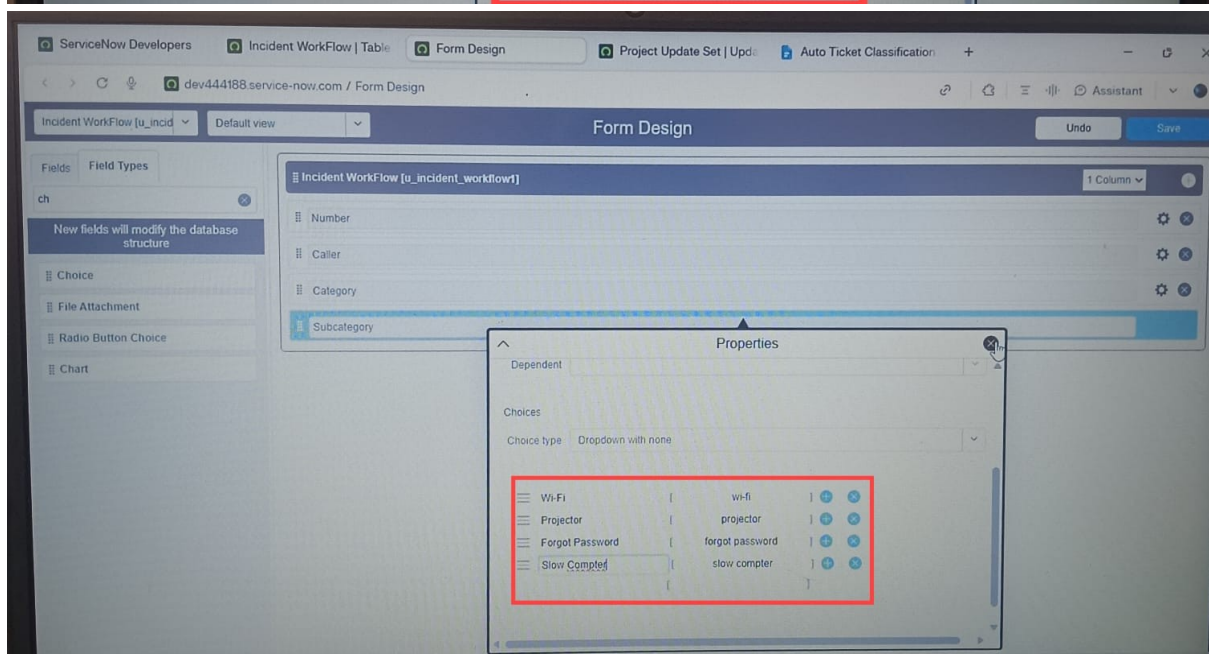
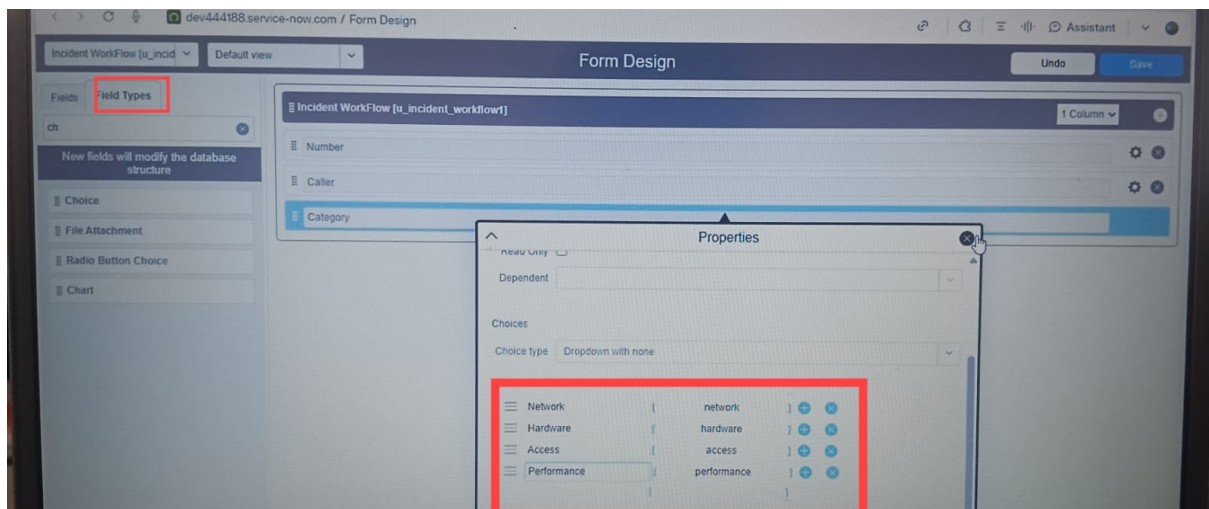


Activity 2: Field Creation and Data Type Configuration

The objective of this milestone is to create all required fields in the School IT Ticket table with correct data types, choices, and dependencies, ensuring structured and consistent data capture.

Field Label	Type	Reference/Choice
Number	Auto Number	—
Caller	Reference	sys_user
Category	Choice	[Network, Hardware, Access, Performance]
Subcategory	Choice	[Wi-Fi, Projector, Forgot Password, Slow Computer]
Short Description	String	—
Description	String	—
State	Choice	[New, In progress, On hold, Resolved, Closed]
Assigned Group	Reference	sys_user_group
Assigned to	Reference	sys_user

- After Creating Fields with their appropriate Type, then add the Choice from the Form design.
- To all the Choice list Fields.
- They are Category, Subcategory and State.



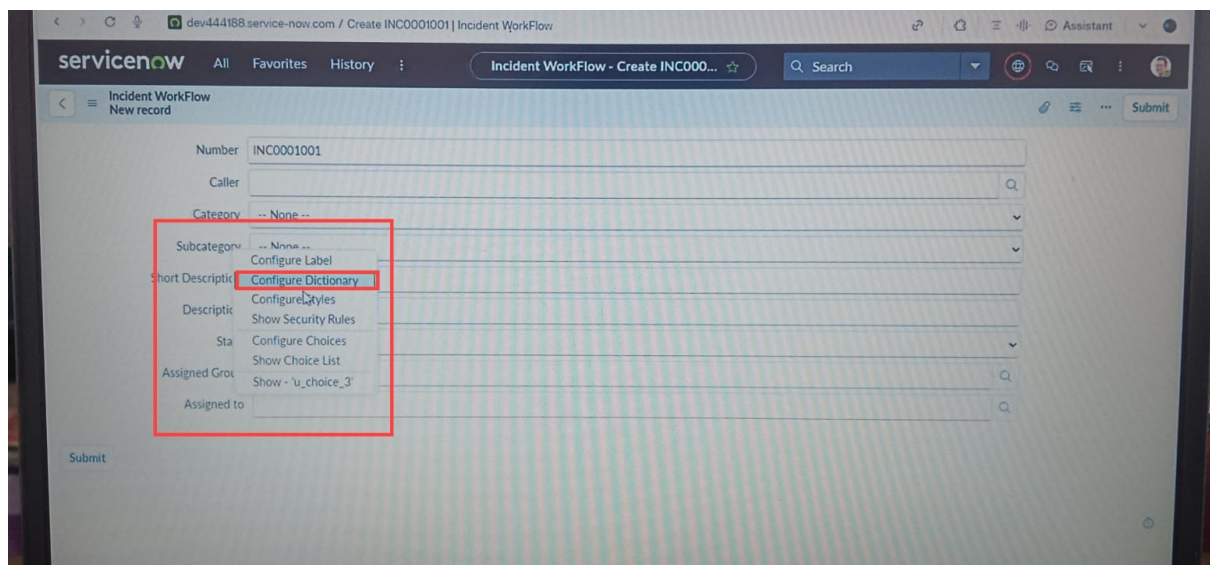
- Same like that, add Choice to the State field.
- Then Click the save button on the top right corner.
- Open the Form and reload it.

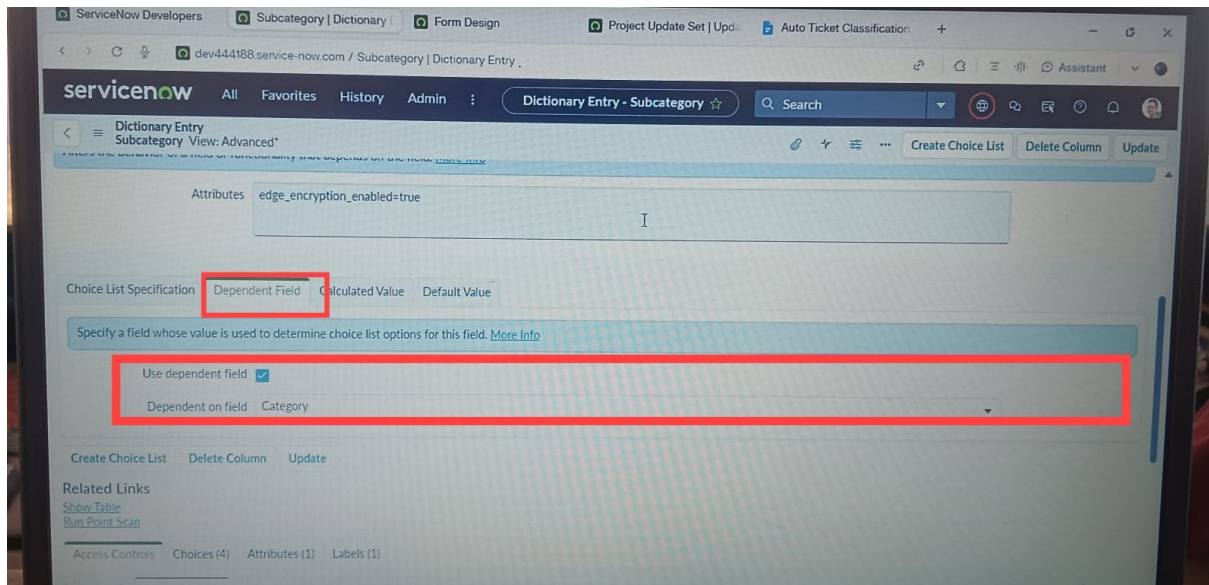
Activity 3 : Implementing Dependency Between Category and Subcategory Choice Fields

The objective of this milestone is to implement a **dependency relationship** between the **Category** and **Subcategory** fields so that the available Subcategory options dynamically change based on the selected Category.

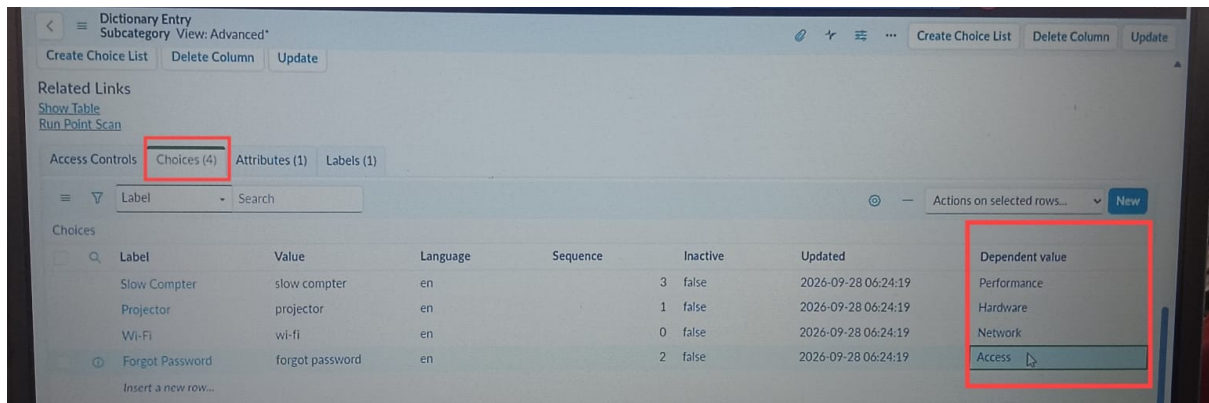
Navigation Steps :

- Open the Created Table Form and Right click on the **Subcategory** Field.
- Click on the **Configure Dictionary**.
- Under **Related links** Click on the **Advanced view**,
- Then select the Dependent **Field** Section.
- Enable the checkbox **Use dependent field true**.
- Then select the Field **Category** on the **Dependent on field**.
- Then **save** the Form.





- Now scroll down to the choice section.
- Then add the dependent fields for which option dependent on the category gives the category choice backend name to the subcategory choice values.
- Wi-Fi → Network.
- Projector → Hardware.
- Forgot password → Access.
- Slow computer → Performance.
- Then save the Form.
- Then it will display each choice to each one separately based on the category subcategory displays.



The screenshot shows the 'Incident WorkFlow' form with the following fields and values:

- Number: INC00509
- State: -- None --
- Assigned group: [Search]
- Assigned to: [Search]
- Category: Access
- Subcategory: -- None --
- Short description: Forgot Password
- Description: [Empty]

Milestone 3 : Automation using Flow Designer & Email Notification.

When a student creates a new IT support ticket, they only need to enter the Caller Name and Short Description of the issue. The student does not need to

manually select the Category or Subcategory. Once the ticket is submitted, a Flow Designer automation is triggered and evaluates the Short Description for predefined keywords such as WiFi, Projector, Password, or Slow. Based on the identified keyword, the flow automatically assigns the appropriate Category and Subcategory to the ticket.

- If the short description contains the word *WiFi*, the flow assigns **Category = Network** and **Subcategory = WiFi**.
- If the description contains *Projector*, the flow assigns **Category = Hardware** and **Subcategory = Projector**.
- If the description includes *Password* or *Login*, the flow assigns **Category = Access** and **Subcategory = Forgot Password**.
- If the description mentions *Slow* or *Hanging*, the flow assigns **Category = Performance** and **Subcategory = Slow Computer**.

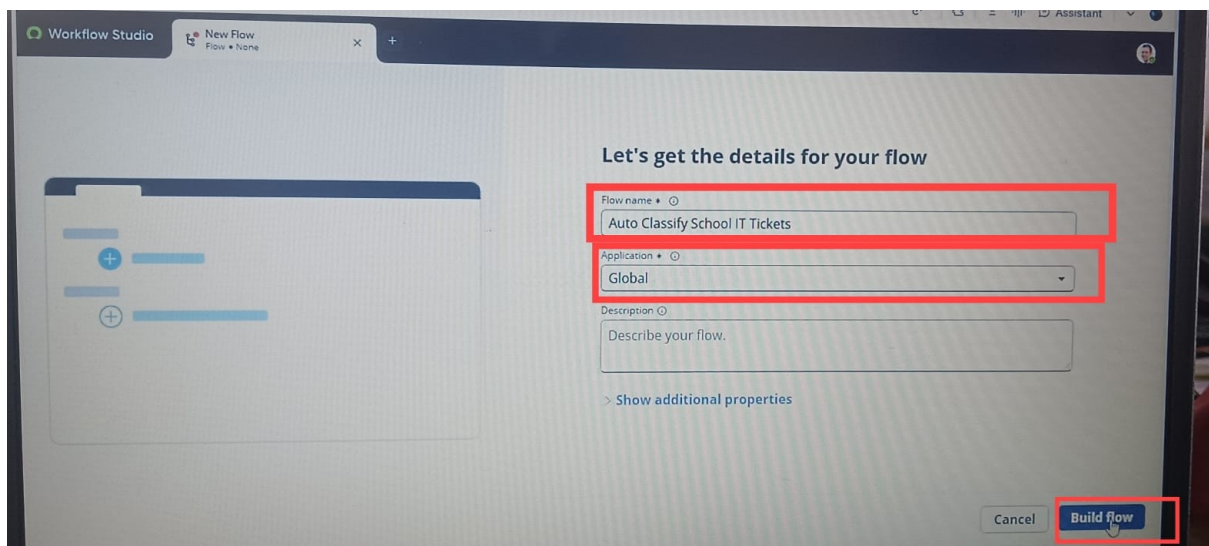
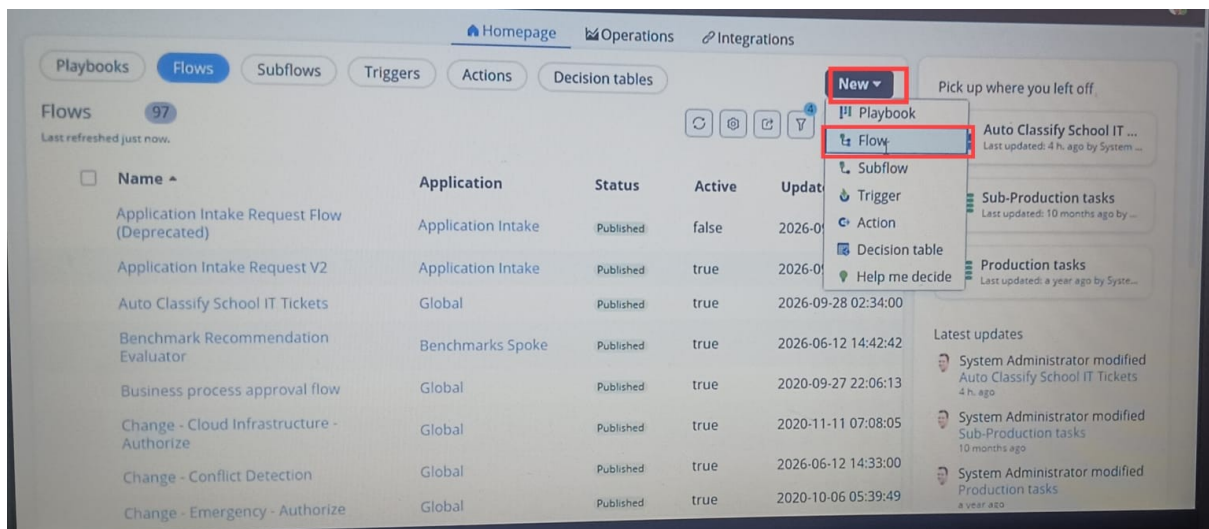
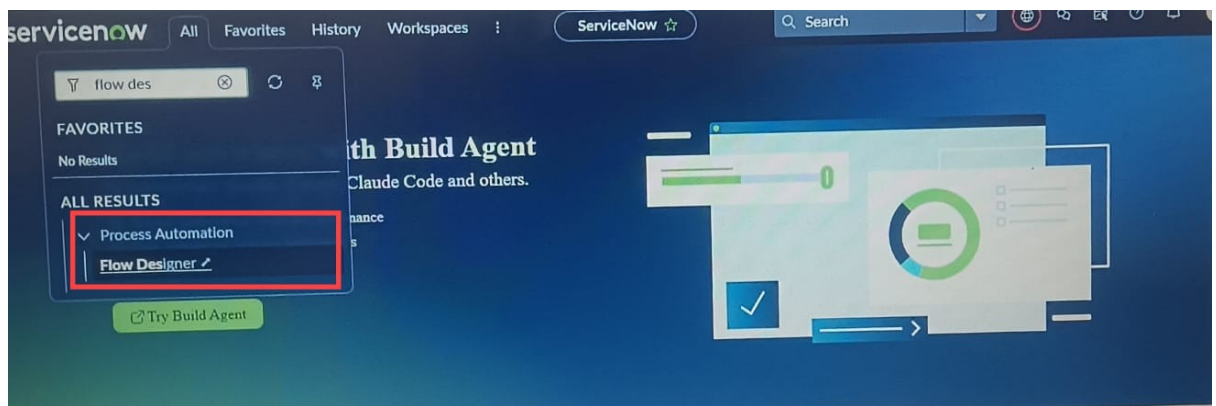
This automation ensures consistent classification, eliminates human error, and significantly reduces the manual effort required by IT staff.

The entire process is implemented using **Flow Designer**, making the solution easy to maintain, scalable, and completely no-code.

Activity : 1 Auto Classification using Flow design.

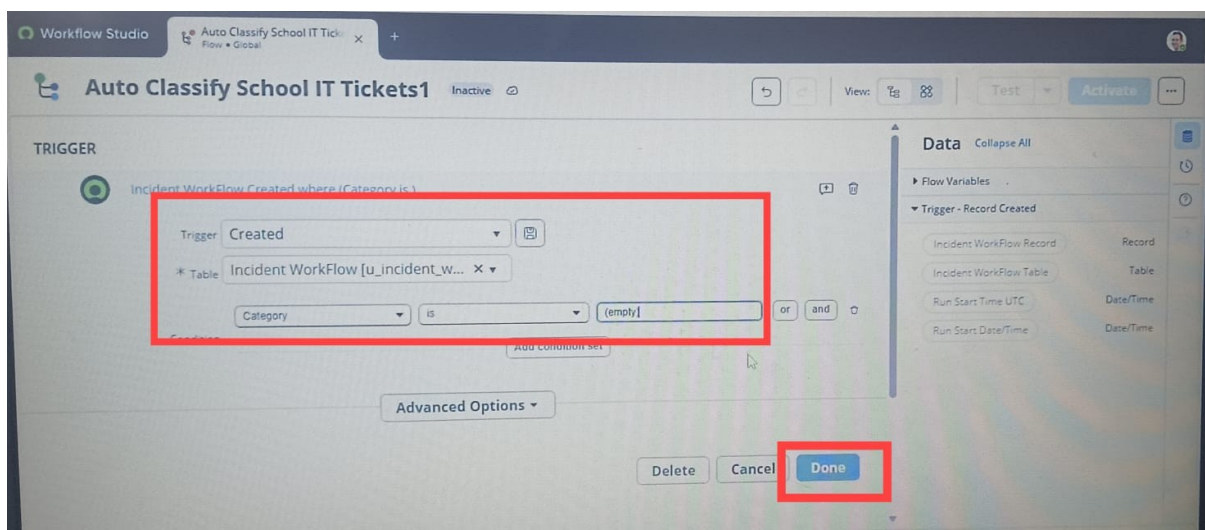
- Go to **All menus** → Search **Flow designer**.
- Under **Process Automation** Open the **Flow designer**.
- On the top right corner Click **New** and Select **Flow** Option.
- Name : **Auto Classify School IT Tickets**.
- Application : **Global**.

- Now Click the **Build Flow** button.



Activity : 2 Trigger Configuration.

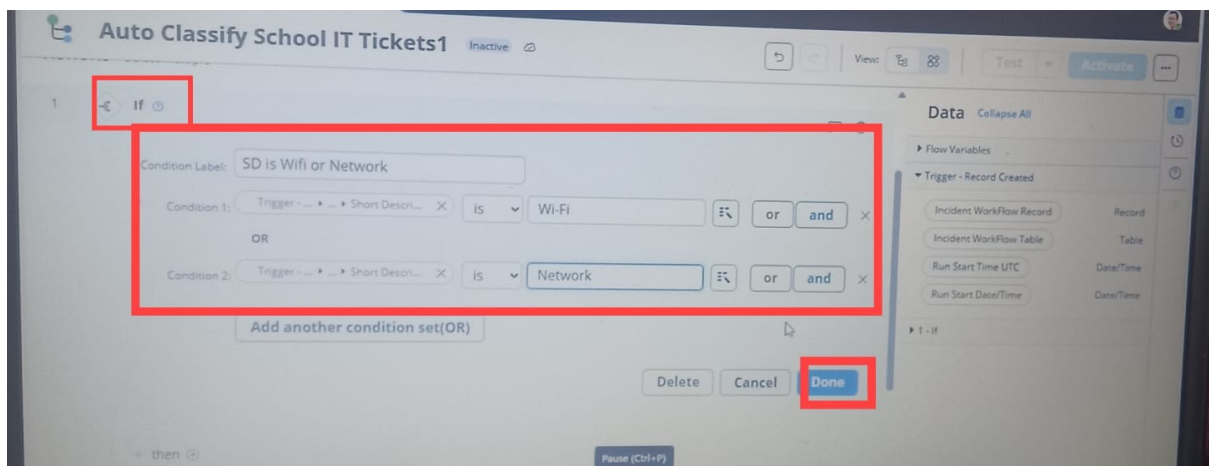
- Trigger : **Record Created.**
- Table : **Incident Workflow.**
- Condition : **Category is Empty.**
- Then Click **Done.**



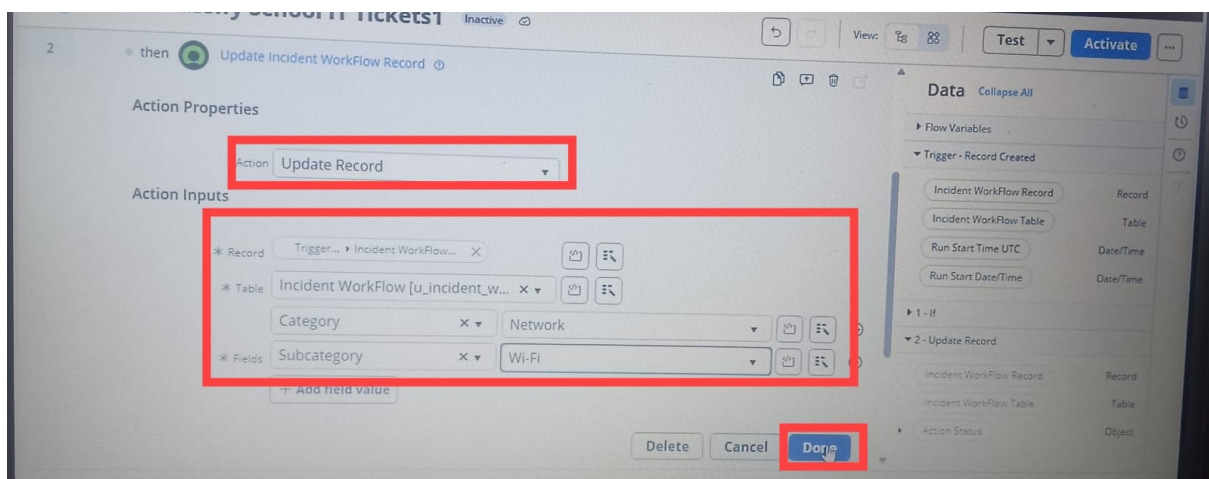
- Now Under **Actions** Select **Flow Logic** Option.

Activity : 3 Wi-Fi Issue Classification.

- Select **If** condition.
- Condition label : SD is **Wifi or Network** (Give as ur wish).
- Condition 1 : Using data pill picker Select **Short description** as Field , and operator **Contains** , Value is **Wi-Fi**.
- Condition 2 : Using data pill picker Select **Short description** as Field, and Operator is **Contains**, Value is **Network**.
- Then Click **Done** Button.



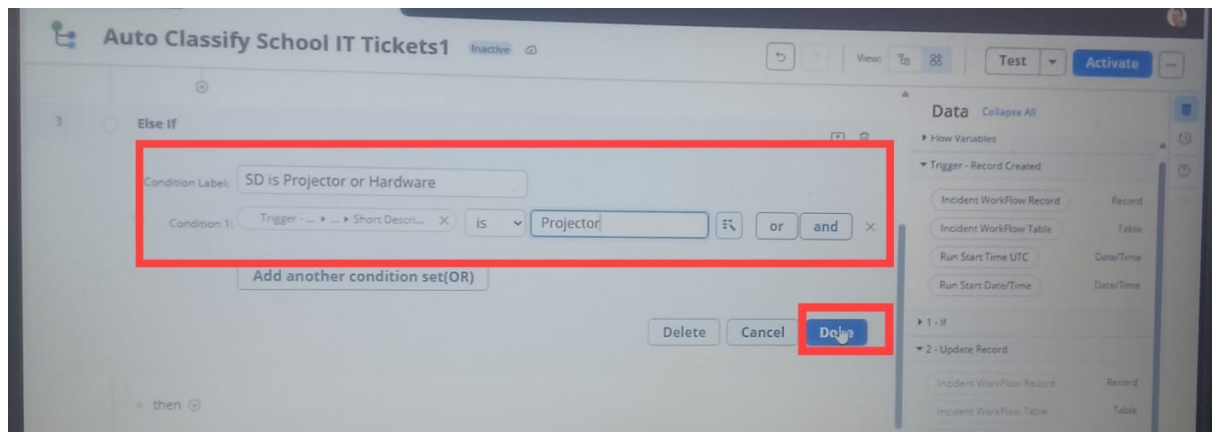
- Now Click **Action** to **Update record**.
- Action Inputs.
- In Record Field drag and drop data from the right side.
- Table : Auto updates as Incident Workflow.
- Fields Set **Category** = **Network**.
- Set **Subcategory** = **Wi-Fi**.
- Then Click the **Done** button.



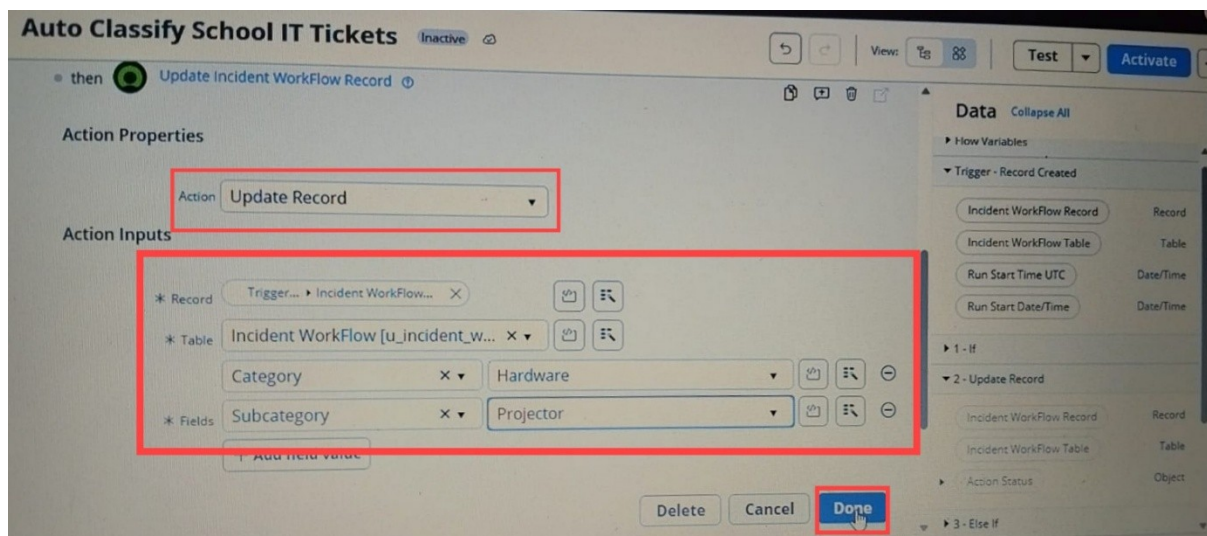
Activity : 4 Projector Issue Classification.

- Now Select the **Else If** Condition.
- Condition label : SD is **Projector or Hardware**.

- Condition : **Short description Contains Projector.**
- Then Click the **Done** button.
- Now Action is an **Update record.**
- Set **Category** = **Hardware.**
- Set **Subcategory** = **Projector.**



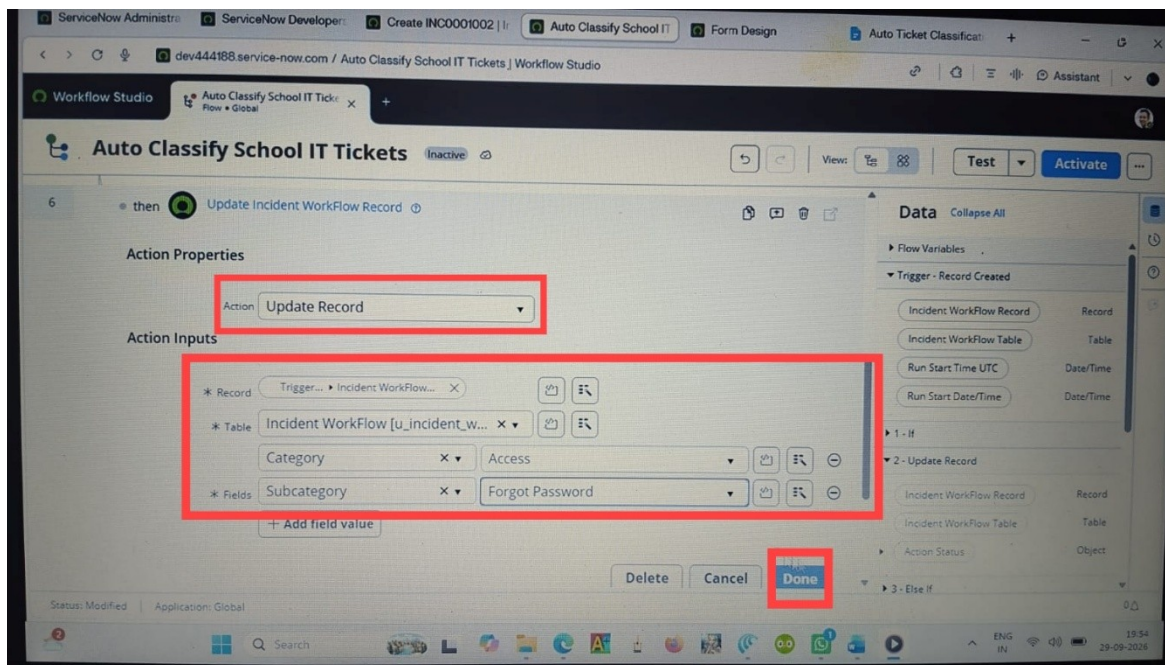
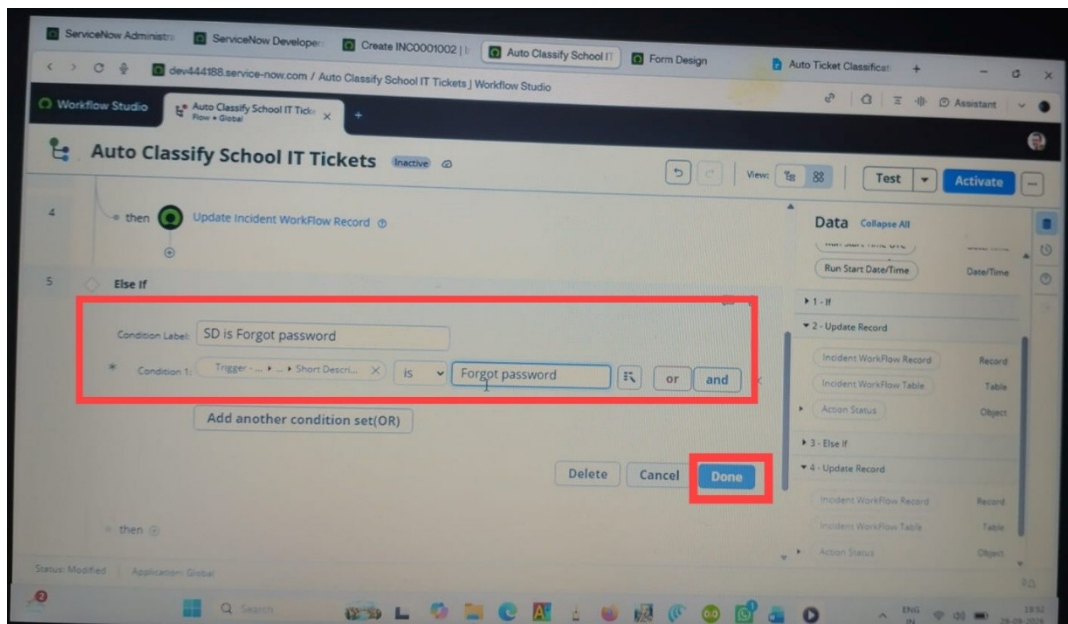
- Then Click the **Done** button.



Activity : 5 Password Issue Classification.

- Now again select **Else If** condition.
- Condition label : SD is **Forgot password.**
- Condition : **Short description contains Forgot password.**
- Then Click the **Done** button.

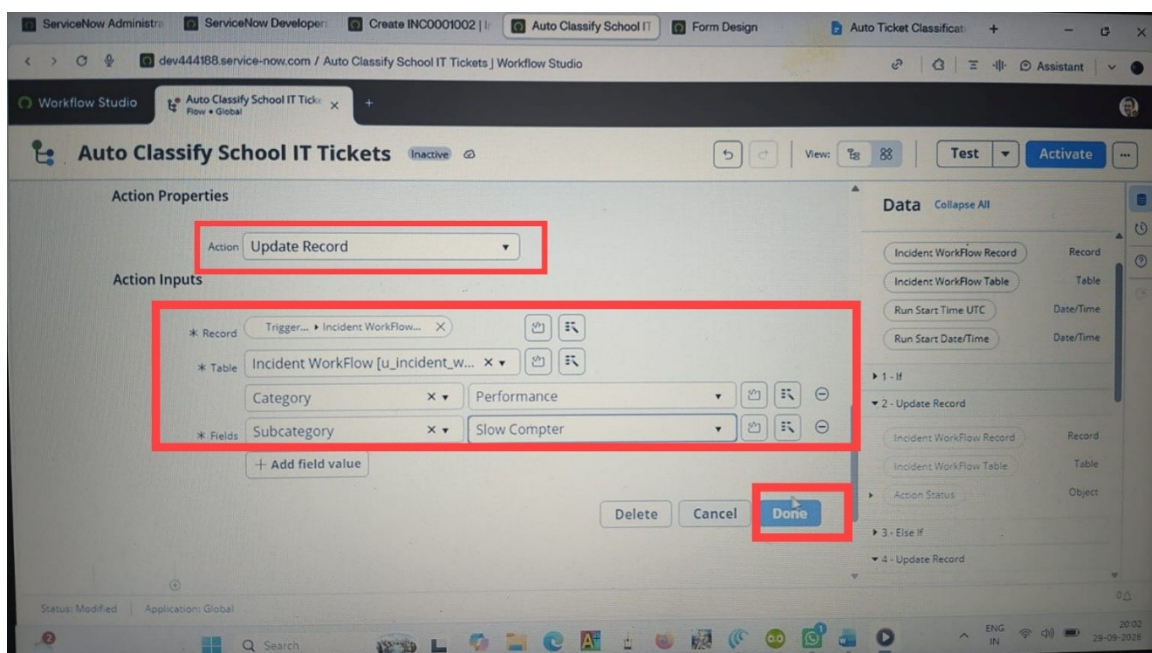
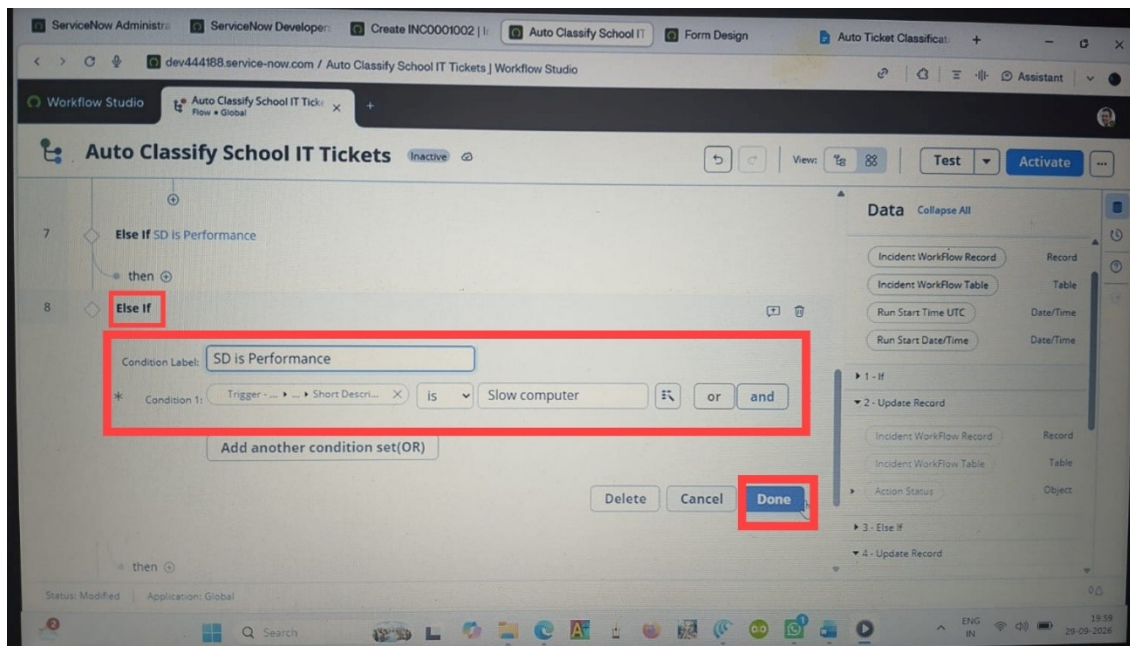
- Now Action is an **Update Record**.
- Set **Category** = **Access**.
- Set **Subcategory** = **Forgot Password**.
- Then Click the **Done** button.



Activity : 6 Slow Computer Classification.

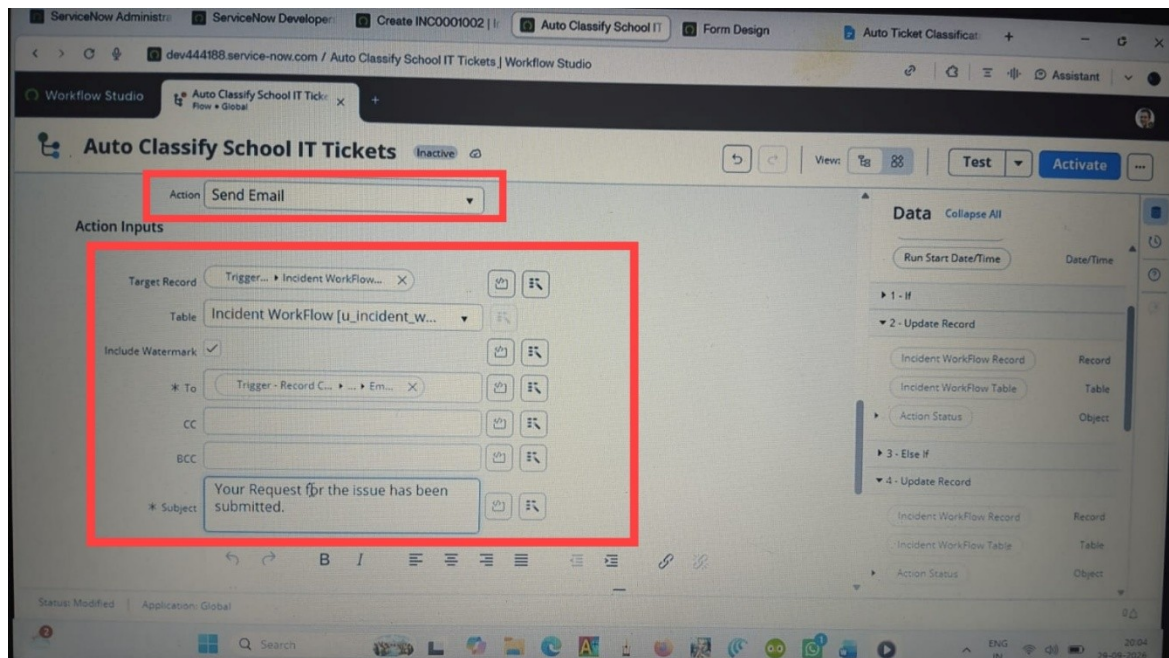
- Now Again Click the **Else If** Condition.
- Condition label : SD is **Performance**.

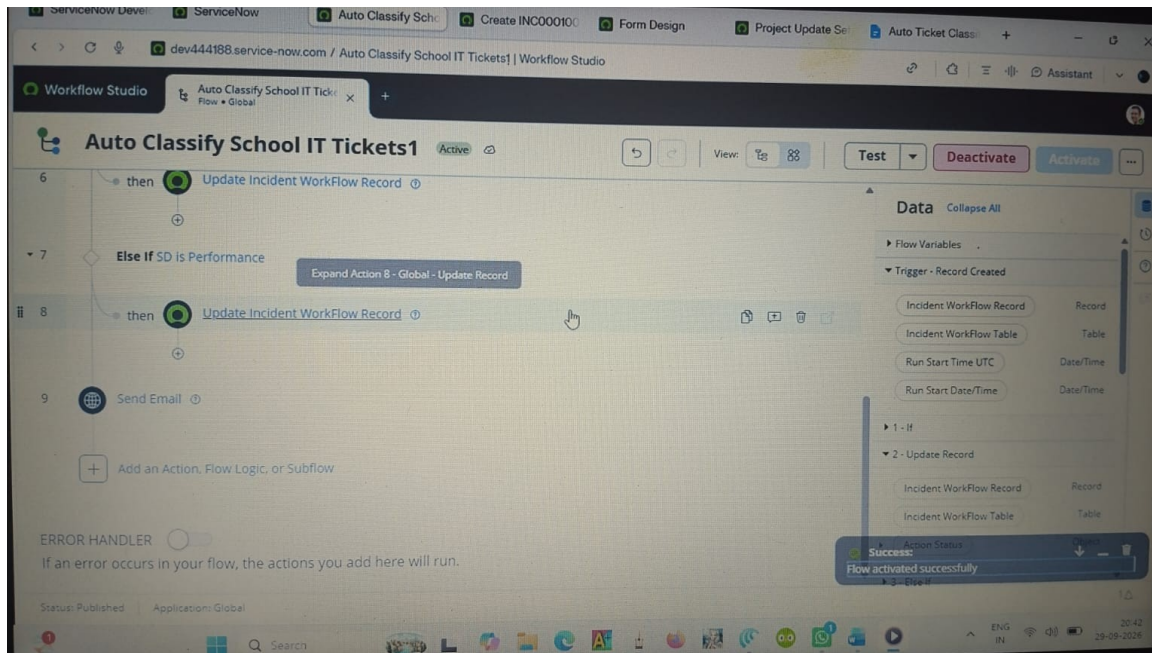
- Condition : **Short description contains Slow Computer.**
- Then Click the **Done** button.
- Now Action is an **Update Record**.
- Set **Category** = **Performance**.
- Set **Subcategory** = **Slow computer**.
- Then Click the **Done** button.



Activity : 7 Email Action Setup.

- After classification, an email notification is sent using Flow Designer.
- Set Action **Send Email**.
- To : **Caller → Email**.
- Subject : **Your Request for the issue has been submitted.**
- Body : **Ticket confirmation message.**
- Then Click the **Done** button.





- Now Click the **Activate** button from the top right corner.

PHASE 6 : Testing, Validation & Security.

- The objective of Phase 4 is to ensure that the developed solution works as expected, meets all functional requirements, and follows basic security and access control best practices.
- This phase validates the end-to-end automation, confirms data accuracy, and ensures that only authorized users can access or modify ticket data.

Data Validation Checks

- Mandatory fields captured correctly.
- Auto-number generated without duplication.
- Category and Subcategory values stored accurately.
- Reference fields resolve correctly to user and group tables.

Activity : 1 Testing & Validation.

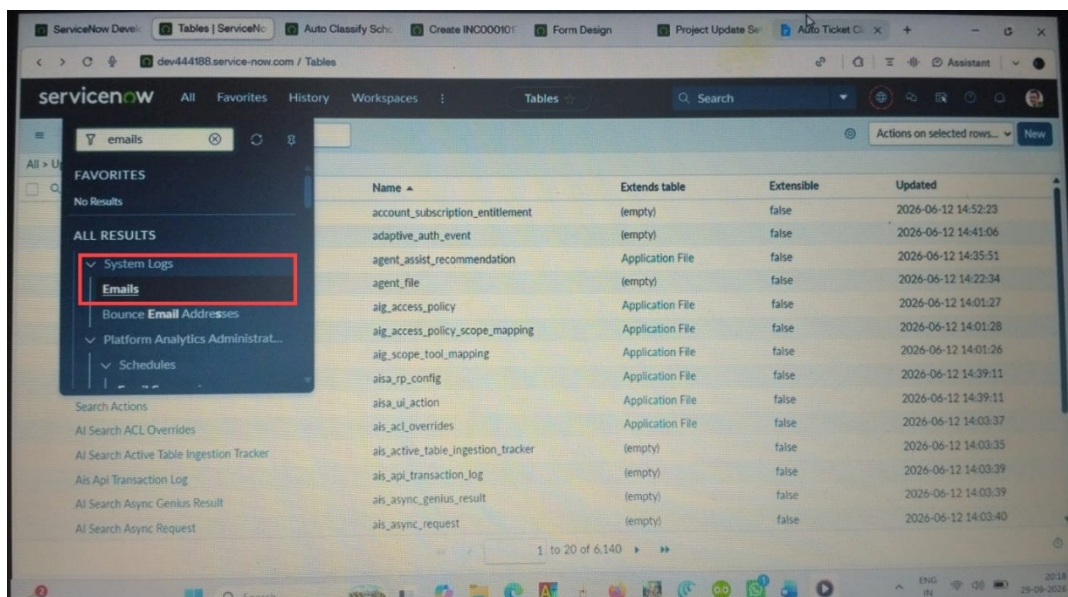
Test Scenario 1

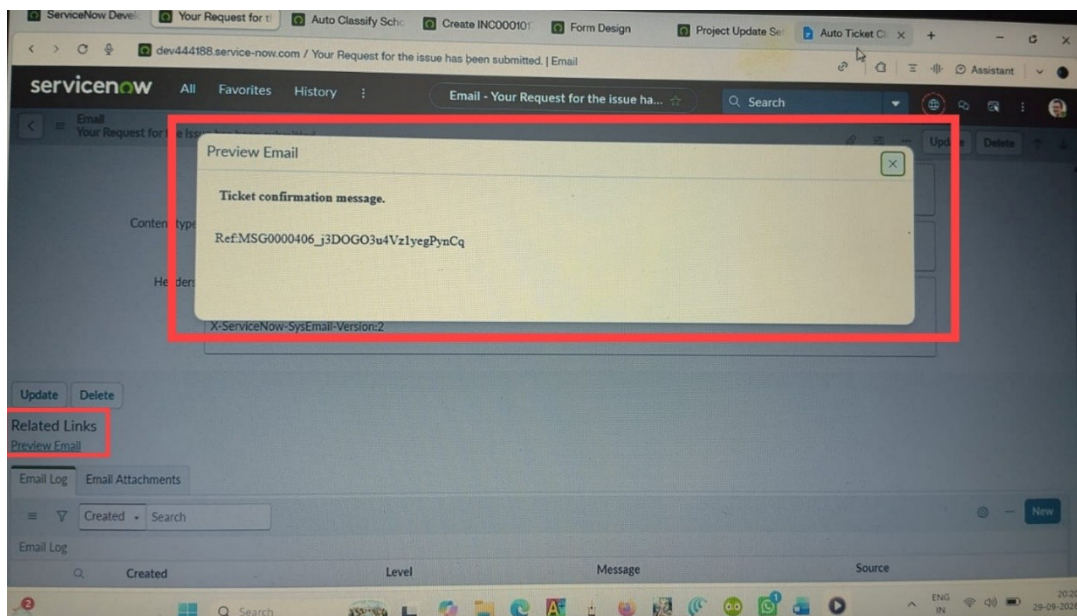
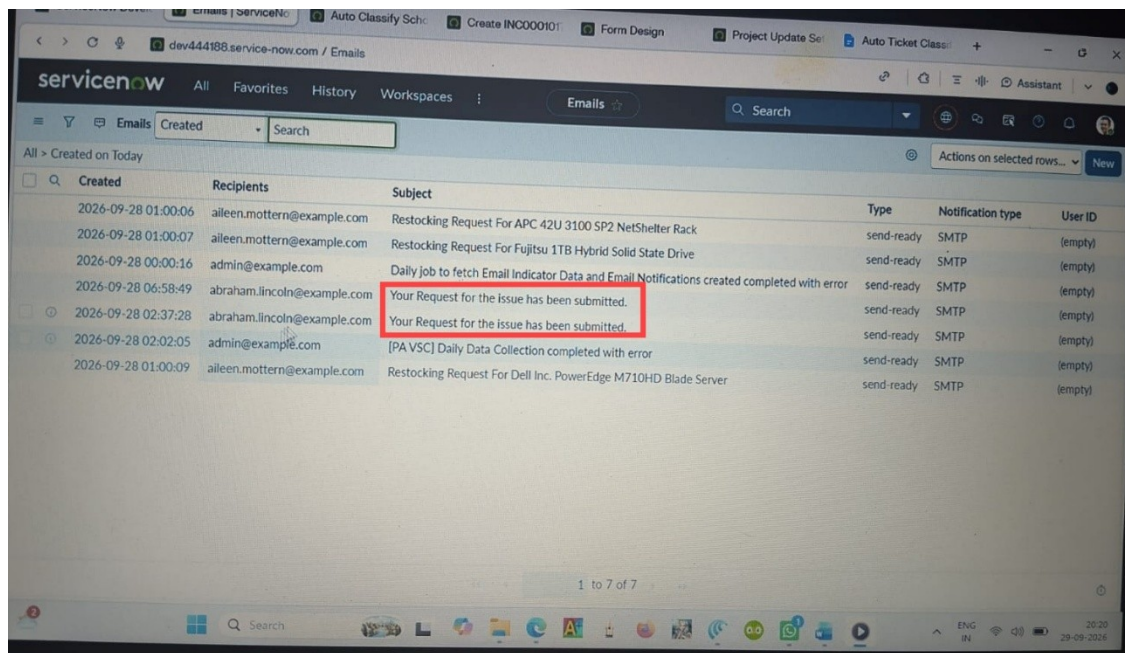
- Enter one Caller Name.

- Short Description: WiFi not working in library
- Then Save the Form.
- And Reload the Form.
- Then it fills auto as :
- Category → **Network**.
- Subcategory → **WiFi**.
- Email sent to caller Email Id.

Activity : 2 Check Email Notification.

- Open All menus → Search Emails.
- Under System logs → Select Emails.
- Search with the Given Subject Name.
- Open that Record and Under Related Links Click Preview mail Option.





Activity : 3 Test Scenario 2

- Enter one Caller Name.
- Short Description: Projector not turning on.
- Then Save the Form.
- And Reload the Form.
- Then it fills auto as :
- Category → **Hardware.**

- Subcategory → **Projector**.
- Then Email sent to caller Mail id.

The screenshot shows the ServiceNow Incident WorkFlow form for INC0001003. The form is titled 'Incident WorkFlow - INC0001003' and has a search bar. The form fields are as follows:

Number	INC0001003
Caller	Abel Triter
Category	Hardware
Subcategory	Projector
Short Description	Projector not turning on
Description	
State	-- None --
Assigned Group	
Assigned to	

Buttons for 'Update' and 'Delete' are visible at the bottom left of the form.

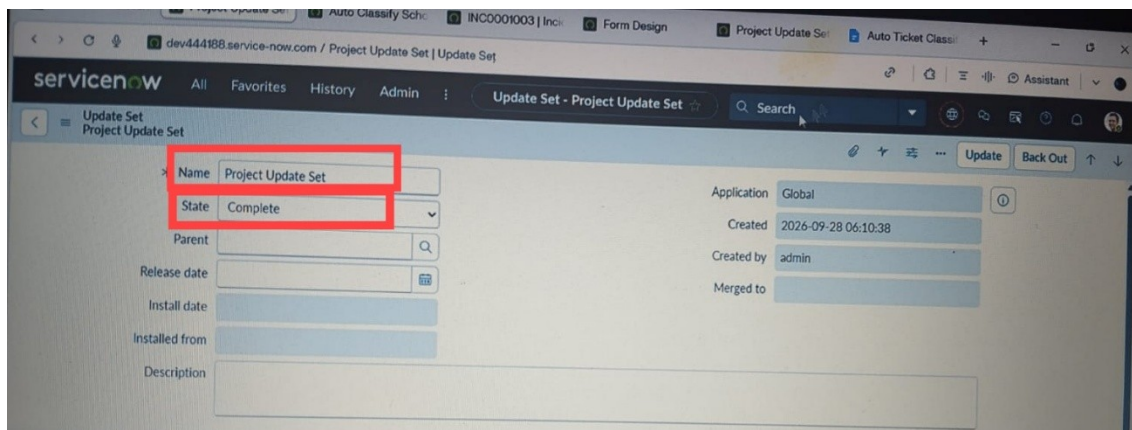
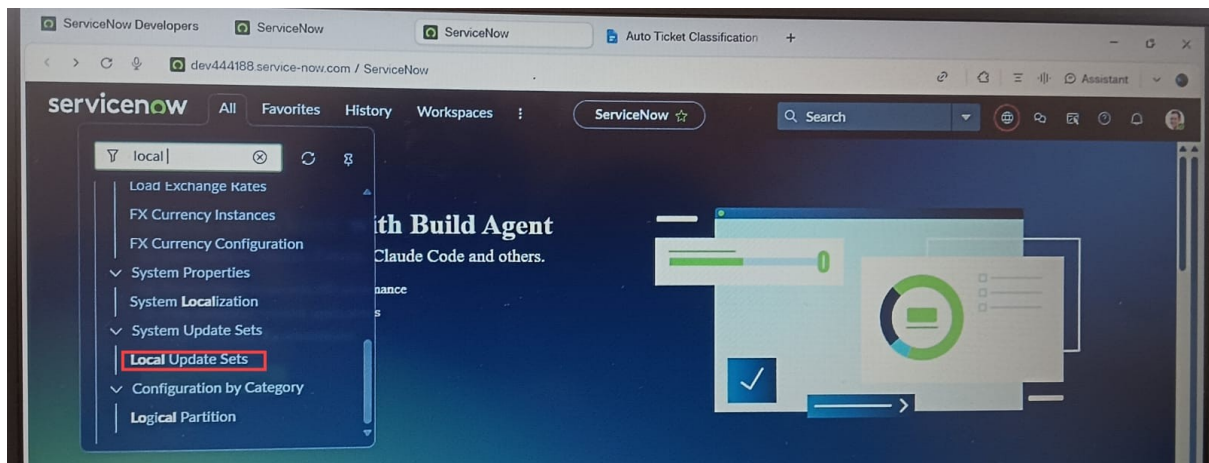
PHASE 5: Deployment & Final Documentation.

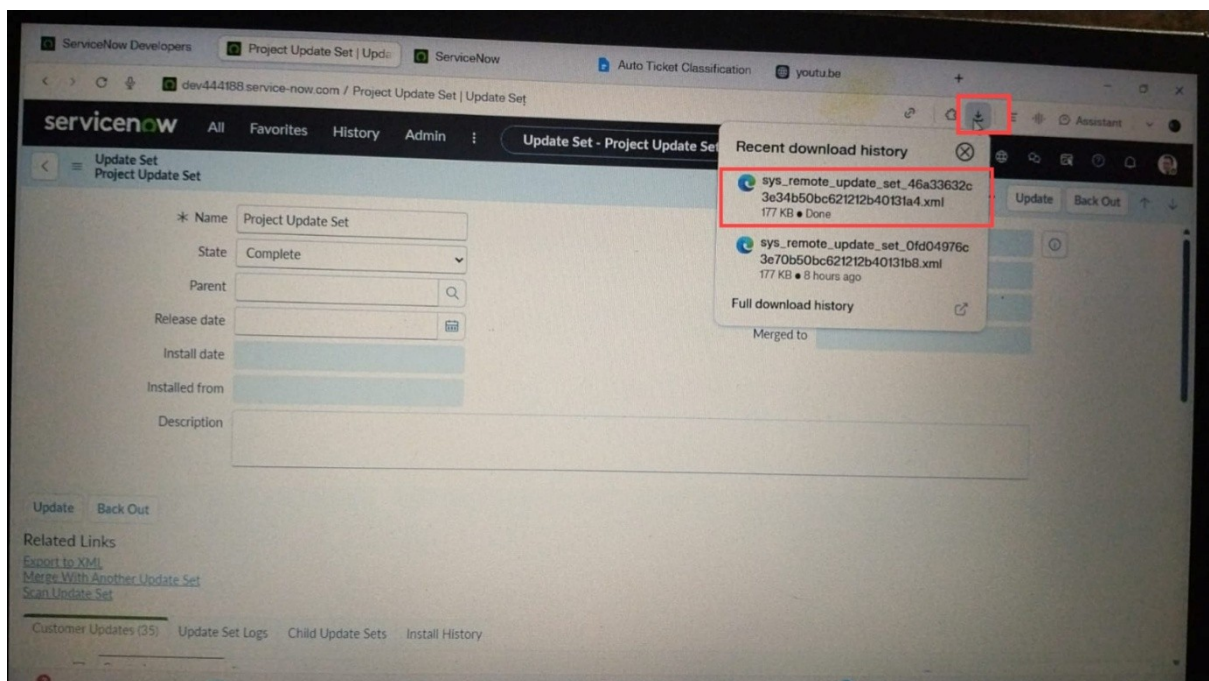
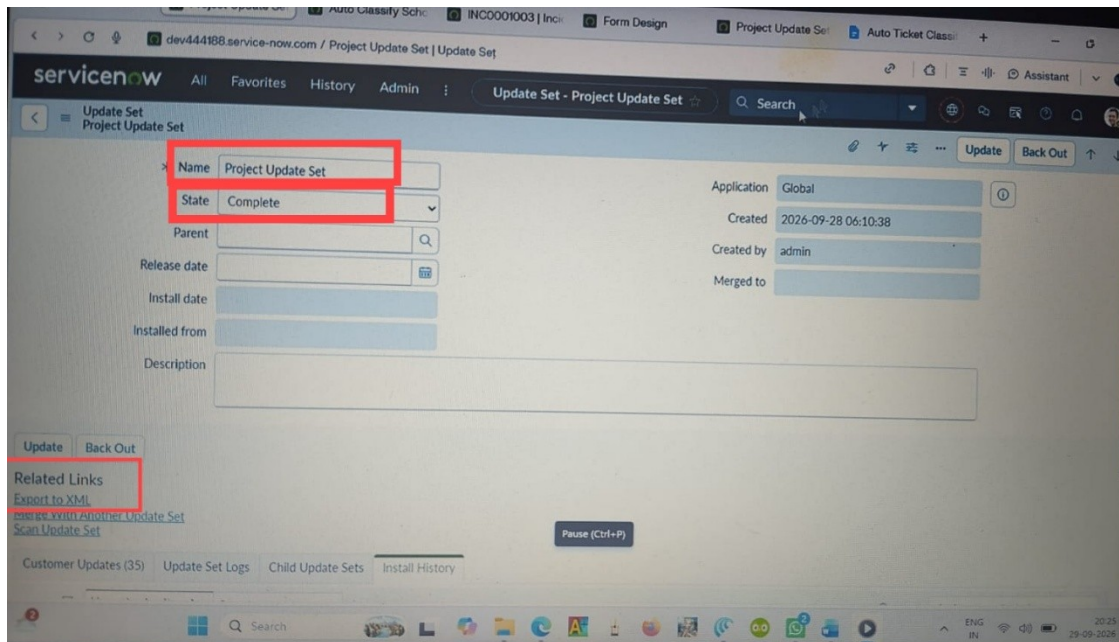
- The objective of Phase 4 is to deploy the developed solution to the target environment and finalize all project documentation.
- This phase ensures that the solution is stable, reusable, and ready for demonstration or production use, along with complete and clear documentation for future reference.

Activity : 1 Make Update set to Complete State.

- Go to All menus → Search For **Local Update Sets**.
- Under **System Update sets** Select → **Local Update sets**.
- Open the **Project Update set**.
- Make the State From **In progress** to **Complete**.
- Then **Save** the Form.

- Under Related links Click **Export to XML** Option.
- Then the Xml Format the Update set data will be **downloaded** to your system.
- We can share through WhatsApp or Mail that downloaded file.
- Then the Created Update set will transform to others in this way.





Conclusion

- The **Auto Ticket Classification using Flow Designer** project successfully delivers a complete, end-to-end automation solution for a school IT helpdesk. The primary objective of reducing manual effort in ticket categorization and improving response efficiency has been fully achieved.
- Through structured requirement analysis, careful backend configuration, and no-code automation using Flow Designer, the system now automatically classifies IT tickets based on the short description provided by students. The implementation of dependent choice fields ensures data accuracy, while automated email notifications improve communication by instantly acknowledging ticket creation to the caller.
- The solution was developed using a phased approach, ensuring clarity, stability, and ease of maintenance. Comprehensive testing and security

validation confirmed that the system functions reliably, maintains data integrity, and enforces appropriate access controls.

- Overall, this project demonstrates how ServiceNow can be effectively used to build scalable, maintainable, and user-friendly automation solutions without relying on complex scripting or machine learning. The design is well-suited for small to medium environments such as educational institutions and can be easily extended in the future to include advanced features like assignment automation, SLA tracking, or predictive intelligence.

Submitted By,

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Santhiya.P

Ramani.K